

KRISH BANSAL

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EDUCATION

Virginia Tech | Blacksburg, VA

Expected May 2027

Bachelor of Science in Computer Science

Cumulative GPA: 3.62 / 4.0 | In Major GPA: 3.68 / 4.0

Dean's List with Distinction: Spring 2025, Fall 2025 | Dean's List: Fall 2023, Fall 2024, Spring 2026

EXPERIENCE

Keshav Enterprises | Intern | India | Python, PostgreSQL, FastAPI, Docker, MLflow

May 2026 – Aug 2026

- Architected end-to-end ELT pipeline in Python and PostgreSQL, normalizing sales inquiry and order data across 4 relational entities to replace manual spreadsheet tracking for an SME manufacturer
- Built a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB vector search and the Claude API to auto-draft grounded customer replies, constraining outputs to retrieved catalog data to eliminate hallucinated pricing
- Designed and ran an ML evaluation harness benchmarking Prophet time-series forecasts against a naive seasonal baseline across 8 product lines, using MLflow to gate model deployment – correctly **rejected the more complex model in 7/8 cases** due to limited (24-month) historical data, preventing an unjustified model from shipping
- Achieved a **49% relative MAPE reduction** (52.0% → 26.6%) with Prophet on the one product line where sufficient trend signal existed, validating the evaluation harness caught a genuine case for deployment

Undergraduate Research Assistant | Dataism Laboratory for Quantitative Finance | Python, R

Oct 2025 – Aug 2026

- Built a CNN-Transformer hybrid model for high-frequency market data, using convolutional filters to extract short-horizon trading signals and transformer attention to model long-range temporal dependencies for price direction forecasting using Python and R

Uber | Intern | Remote

Oct 2022 – Dec 2022

- Performed data-driven market analysis and competitive benchmarking to identify user segments and feature opportunities for the “Uber-Teens” prototype, and designed high-fidelity UI/UX mockups, optimizing user flows and interaction design while presenting product strategy insights to Uber stakeholders and industry mentors

PROJECTS

Change Assurance Runtime for AI Coding Agents | VT Hacks 14 | Python, FastAPI, Textual, Electron/React, SQLite | 🌟 Sep 2026

- Built a local first Windows runtime that puts AI coding agents under independent supervision, using kill on close Job Objects and restricted token process launches to attribute every descendant process by PID, image path, and command line
- Designed a tamper evident, hash chained event journal with end to end replay verification and a signed Ed25519 Change Passport capturing intent, authority, evidence, and outcomes for every agent action
- Shipped three clients (Textual TUI, CLI, Electron/React desktop) against one frozen OpenAPI contract of 66 operations, 61 routes, and 110 typed schemas, validated by a threat model review (15 of 16 findings fixed) and 1,022 passing tests

Regime Detection & Adaptive Trading Strategies | Python, pandas, NumPy, hmmlearn, scikit-learn | 🌟 Mar 2026 – May 2026

- Built a Gaussian Hidden Markov Model (HMM) in Python to detect market regimes from SPY/VIX data using volatility, momentum, liquidity, and volume features
- Developed a transaction-cost-aware backtesting framework with filtered probability inference and benchmarked against K-Means and momentum strategies; trained on 2015–2020 and tested out of sample on Jan 2021 – May 2026 (5 bps costs), achieving a 57.97% return (Sharpe 0.83) with ~34% lower volatility and a –15.3% max drawdown vs. –24.5% for Buy-and-Hold

Fork-Join Thread Pool | CS 3214 | C, pthreads, Helgrind, Git

Spring 2026

- Built a work-stealing fork/join thread pool in C with pthreads, evolving a single-lock work-sharing queue into per-worker dequeues with finer-grained locking so exactly one thread runs each task
- Prevented deadlock by having workers run not-yet-started joined tasks inline, kept idle workers blocked on condition variables, checked for data races with Helgrind, and benchmarked parallel speedup on 32-core Xeon machines

Real Estate Analyzer | VT Hacks 12 | React.js, Node.js, PostgreSQL, Azure ML Studio

Oct 2024 – Nov 2024

- Built a Boosted Decision Tree model predicting real estate prices with $R^2 = 0.975$ on 428K+ listings across 45+ U.S. states, reducing MAE to \$87.9K and RMSE to \$181.8K via hyperparameter tuning and feature engineering in Azure ML Studio
- Delivered a full-stack web prototype at VTHacks 12, integrating the model with React.js, Node.js, and PostgreSQL to enable scalable real estate price analysis

Playlist Recommendation | Java, SQL

Feb 2024 – May 2024

- Engineered a personalized playlist recommendation system in Java using hybrid filtering, processing 10K+ songs and 1K+ user preference profiles to generate ranked recommendations
- Improved recommendation relevance through dynamic weighting and similarity scoring, reducing average selection time and increasing user engagement

Inventory Management System | Java, SQL

Aug 2023 – Oct 2023

- Architected and implemented a scalable inventory management system in Java with a relational SQL backend, supporting real-time stock tracking, automated restocking logic, and invoice generation
- Designed normalized database schemas and modular object-oriented architecture (OOP), optimizing query performance and ensuring data integrity while documenting system design and scalability in a detailed Technical Design Document (TDD)

TECHNICAL SKILLS

Languages: Python, C, Java, SQL, TypeScript

ML & Frameworks: PyTorch, scikit-learn, pandas, FastAPI, React

Databases: PostgreSQL, ChromaDB

Tools & Cloud: Docker, MLflow, Git, Google Cloud Platform